



สถาบันวิจัยเพื่อการพัฒนา (IRD)



จุฬาลงกรณ์มหาวิทยาลัย



มหาวิทยาลัยเทคโนโลยีราชมงคลล้านนา น่าน



วิทยาลัยชุมชนน่าน



มหาวิทยาลัยเกษตรศาสตร์

Agroecology BSc Curriculum Development

ADAPT Workshop Report

February 10 and 25, 2026

Based on the results of the previous workshop, held in December 2025, we further discussed the content of the BSc curriculum and key pedagogical orientations as summarized in the report below and reflected in the new version of the curriculum as of February 2026.

Courses content and organization

1. The 4 hours lessons initially planned for after the field block of November 2026 will be moved before the field visit and incorporated into the first 4 weeks of training.
2. The 4 weeks of training prior to the field visit are organized as 4 hours courses organized as 3 sessions each for a total of 16 hours. They combine distant teaching with video recorded courses that introduce agroecology concepts and clip videos that present the situation in Nan Province.
3. A description of the courses is provided below and in the curriculum document:

Week 1: Foundations of Agroecology (4 hours – Aj. Buncha KU)

Week 2: Sustainable Agriculture and Agroecology Practices (4 hours –Aj. Priyanut RMUTL-Nan)

Week 3: Systems Thinking and Collaborative Learning (4 hours - Aj. Pongchai CU-FS)

Week 4: Community Based Solutions and Agroecology Food Systems (4 hours - Aj. Thansiporn CU-IA)

During the follow-up online meeting of February 25, we discussed in detail the curriculum content of weeks 1 & 2. The content of weeks 3 & 4 will be discussed through dedicated group meetings later on. Working group appointment letter will be sent to each member's home institution. The details of responsibilities on video lectures and illustration clips are provided in the curriculum draft document.

Field block in Nan

During the workshop, we decided to organize the 7-days field block in Nan Province during the week of 16 to 20 November 2026 as it meets the requirements of all partners involved.

The field block will be organized by all partners under the coordination of Dr. Peng of Nan CC, Ajan Priyanuth of RMUTL-Nan and Ajan Thansiporn of CU-IA. A field visit of the Nan team will

be organized during the last weekend of March 2026 to finalize the program of the field block. The first 2 days of the field block, dedicated to rice systems, may be organized in Tha Wang Pha District with a night spent at the Silaphet Rice Farmer School.

Video production team and e-learning system

We set up a video design team composed of Aj. Priyanuth, Aj. Jean, Ms Sirinan and Ms Nichaphart. This team will coordinate with a Nan video company the production of eight video clips to illustrate different points of the curriculum: gaming and simulations, animal raising (red cows and pig semen), fermented liquid fertilizer, learning centers, etc. In addition, teaching content will be recorded using online software, supplemented by existing short video clips. Video production may also be integrated into the curriculum as a student activity — students will create and present short clips on assigned topics, which may be entered into a competition.

An online learning platform will host the different teaching materials. It may be based on **LMS (Learning Management System)** at Kasetsart University: Video-based online learning with attendance tracking. Students must complete each video before proceeding to the next week's content. Students can submit questions to instructors, followed by a live joint session. Aj. Buncha will liaise with Kasetsart University regarding the central system setup. If the system is not ready or proves too complex, regular online learning will be used in the interim. **MyCourseville** the Chulalongkorn University platform will also be investigated by the project team. Other popular online learning platforms currently in use: **MOOC, MS Teams, Zoom**, etc. Each module includes a post-learning exercise that students must pass before proceeding to the next topic, thereby increasing learner engagement with the content.

Issues to consider in teaching and knowledge delivery

1. Internet connectivity problems in Nan Campus.
2. Uncertainty regarding the location of Chulalongkorn University's School of Agricultural Resources.
3. Teaching schedules must be adapted to circumstances, including natural disasters and local conditions.
4. Cost considerations for course delivery, since each university operates on different academic calendars — particularly for non-credit enrollment.
5. Prepare evaluation framework with a dedicated team.



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Agroecology and Sustainable Agriculture: Field-Based Learning in Nan Province

Collaborative Curriculum 45 Hours | 3 Credits | 7-Day Intensive Field Block

PART A: COURSE FOUNDATION

Course Overview

Title: Agroecology Experiential Learning in Nan Province

Level: Upper-level undergraduate (BSc)

Partner Institutions:

- **Kasetsart University (KU)** Faculty of Agriculture
- **Chulalongkorn University (CU)** integrates:
 - Faculty of Science - Companion modelling and role play games
 - SAR - Community-based solutions and business models
 - CLNR - Practical training and coordination
- **Rajamangala University of Technology Lanna - Nan (RMUTL-Nan)** - Local context & case studies
- **Nan Community College (NCC)** - Field site coordination and community partnerships
- **ADAPT-IRD Project** - Pedagogical framework and quality assurance

Delivery Model: Hybrid

- Pre-field seminars (16 hours): At home institutions or online
- Intensive field block (29 hours): 7 days in Nan Province

Total Hours: 45 hours (3 credits)

Target Students:

- Agriculture, environmental studies, sustainable development students
- Community college students in sustainability programs
- Extension workers and agricultural practitioners
- Early career professionals seeking agroecology training

Learning Outcomes

Upon completion, students will be able to:

1. **Apply agroecological principles** to analyze agricultural systems at multiple scales (farm, landscape, food system)
2. **Practice systems thinking** to understand interconnections between ecological, social, economic, and political dimensions of agriculture

3. **Engage with farmer knowledge** respectfully and reciprocally, recognizing farmers as knowledge holders and innovators
4. **Facilitate participatory processes** including companion modelling and role play games for agroecosystem analysis and community-based research
5. **Evaluate sustainability** across environmental, economic, and social dimensions using integrated assessment approaches
6. **Design agroecological solutions** that address real challenges in specific contexts, **connecting local practices** to broader food system dynamics, markets, and policies

PART B: CURRICULUM STRUCTURE AND CONTENT

Phase 1: Pre-Field Preparation (16 hours over 4 weeks)

Delivery: Online synchronous sessions and/or self-paced modules

Week 1: Foundations of Agroecology (4 hours)

- **Pre-reading:** Selected chapters on agroecology fundamentals
- **Assignment:** Students write reflections on agroecological principles and how they relate to sustainable development goals in Nan Province or in agricultural systems they know (2 pages)

Session 1: Presentation of the Agroecology Curriculum and Learning Assessment (1 hour)

- **Lead:** Kasetsart University, Chulalongkorn University, and Nan Community College
- **Content:**
 - Introduction to experiential learning and transformative pedagogy
 - Knowledge acquisition and learning evaluation process
- **Learning Objectives:**
 - Intentions of the curriculum and expectation from the trainers in terms of pedagogy
 - Beyond knowledge, capacity building toward stewardship, leadership, coaching and facilitation of participatory processes
- **Teaching modules**
 - 1.1 Introduction to the curriculum on agroecology and learning journey (VDO PPT 15 min - Aj. Buncha and Aj. Pongchai)
 - 1.2 The assessment process - beyond knowledge, engagement: curiosity, collaboration, citizenship (VDO PPT 15 min – Dr. Peng and Aj. Jean)

Session 2: From Ecological Principles in Agriculture to Agroecology Principles (1.5 hours)

- **Lead:** Kasetsart University
- **Content:**
 - Agroecosystem functioning (farm to global) and dynamics of socio-ecological systems

- Biodiversity functions, nutrient cycling and energy flows in agricultural systems
- **Learning Objectives:**
 - Understand basic ecological concepts; Apply ecological thinking to agricultural systems; Introduction to Nan Province agroecological context
 - Analyze multi-scale agroecosystem functioning; Understand biodiversity's role in agricultural systems; Recognize patterns and processes in agroecosystems
- **Teaching modules**
 - 2.1 Functioning of agroecosystems, from farm to global (VDO PPT 10 min - Aj. Buncha)
 - 2.2 Biodiversity and nutrient cycling (VDO PPT 10 min - Aj. Buncha)
 - 2.3 Energy flows in agricultural systems (VDO PPT 10 min - Aj. Parichat)
 - 2.4 Intro/guidance Nan agricultural context (VDO PPT 5 min - Aj. Sakarin), followed by VDO "Raks Pa Nan" (RMUTL-Nan NCC to select)
- **Assessment:** Essay and quiz (3–5 questions per sub-module except 2.4)

Session 3: From Agroecosystem Transformations to Agroecological Transitions (1.5 hours)

- **Lead:** Kasetsart University; **Field Study:** KU + NCC
- **Content:**
 - Climate change impacts on agriculture and food systems; socioecological changes enlarge the scope of agroecology
 - Consider social and cultural dimensions of agriculture and food system transformations
- **Learning Objectives:**
 - Understanding the green revolution, agrarian transition, and environmental degradations
 - Connect production to consumption; Understand global and local food systems; Analyze food security dimensions; Consumer issues
- **Teaching modules**
 - 3.1 Lessons of the Green Revolution (VDO PPT 10 min - Aj. Buncha)
 - 3.2 Climate change overview (VDO PPT 10 min - Aj. Witsanu)
 - 3.3 Climate change impacts on agricultural systems (VDO PPT 10 min - Aj. Parichat)
 - 3.4 Social and cultural dimensions of change (VDO PPT 10 min - Aj. Oranatda)
 - 3.5 Case study from Nan Province – community adaptation to changes (8 min, 1 or 2 clips by video team — Aj. Priyanut)
- **Assessment:** Quiz (3–5 questions per module)

Week 2: Sustainable Agriculture and Agroecology Practices (4 hours)

- **Pre-reading:** Selected documents and videos on sustainable agriculture and agroecological practices; Nan Context: geography, climate, agricultural history; Policy landscape for sustainable agriculture and agroecology movement in Nan Province

- **Assignment:** Students write reflections on a selected agroecological practice and how it relates to agroecological principles, based on a case study in Nan or elsewhere (2 pages)

Session 4: Sustainable Agriculture Approaches and Livelihood Diversification (2 hours)

- **Lead:** Kasetsart University + RMUTL-Nan
- **Content:**
 - Sustainable development concepts, organic agriculture principles and natural agriculture practices
 - Guest presentation: Video from Nan practitioners, e.g. Kul Panyawong (Upstream Nan Community Natural Agriculture Center)
- **Learning Objectives:**
 - Compare different sustainable agriculture approaches; Understand livelihood strategies; Move beyond conventional business or optimization thinking
 - Design sustainable agroecosystem management strategies; Balance production and conservation
- **Teaching modules**
 - 4.1 Sustainable development concept (VDO PPT 10 min - Aj. Buncha)
 - 4.2 Organic agriculture (VDO PPT 10 min - Aj. Supot)
 - 4.3 Natural agriculture (VDO PPT 10 min - Aj. Supot)
 - 4.4 Introduction to organic and sustainable agriculture in Nan Province (VDO introduction 5 min - Aj. Priyanut + VDO clips: Royal Project Learning Center, Lemon Farm, etc.)
- **Assessment:** Collective open discussion/debate in respective universities

Session 5: Farming practices for Sustainable Soil and Water Management (1 hour)

- **Lead:** CU-SAR
- **Content:**
 - Challenges of soil degradation and sustainable soil management, Water resource protection
 - Crop rotations, intercropping, agroforestry systems; Resource efficiency: recycling nutrients, using local renewable resources
- **Learning Objectives:**
 - Diagnosing soil and water problems
 - Agroecological practices and sustainable management of natural resources
- **Teaching modules**
 - 5.1 Heavy metals and chemicals (VDO PPT 10 min Aj. Thansiporn / Aj. Supin)
 - 5.2 Soil and water (VDO PPT 10 min, Aj. Thansiporn / Aj. Thitipat)
 - 5.3 Biochar (VDO PPT 10 min, Aj. Supin)

Session 6: Farming practices for Sustainable Pest Management (1 hour)

- **Lead:** Kasetsart University
- **Content:**
 - Challenges of use and abuse of chemical for pest control
 - Integrated Pest Management (IPM), from substitution of chemicals by biopesticides to redesign of agricultural landscapes
- **Learning Objectives:**
 - Understand agroecological crop protection, Alternatives to chemical pest management;
 - IPM principles and practices; Pathways to reduce chemical dependence
- **Teaching modules**
 - 6.1 IPM (VDO PPT 10 min, Aj. Supot / Aj. Pimsiri)
 - 6.2 Crop rotation (VDO PPT 10 min, Aj. Parichat / Aj. Pimsiri)
 - 6.3 Natural enemies and beneficials (VDO PPT 10 min, Dr. Kittipum)

Week 3: Systems Thinking and Collaborative Learning (4 hours)

- **Pre-reading:** Selected articles and videos on socioecological systems, wicked problems and companion modeling approaches
- **Assignment:** Students integrate knowledge from the previous sessions across topics; Prepare mental maps of Nan agriculture and articulate their expectations/questions

Session 7: Systems thinking for complex systems and "wicked problems" (1.5 hours)

- **Lead:** CU + RMUTL-Nan
- **Content:**
 - Students engage in group discussions about agroecology challenges in Nan Province, through online or local facilitation support.
 - They collectively simulate a multi-stakeholder negotiation as the participants play different roles: farmers, academics, practitioners, traders, administrations, etc.
- **Learning Objectives:**
 - Develop research skills; Facilitate participatory processes; Apply systems thinking on codesigning research projects on real agroecological issues in Nan Province
 - Experiment participatory processes in research and analysis, using online tools for collective mind mapping

Session 8: Companion Modelling (1.5 hours)

- **Lead:** CU + NCC
- **Content:**

- Systems thinking: Trajectories, transitions and transformations; Feedback loops and emergent properties; "Wicked problems" in agriculture and natural resource management
- Participatory methods and companion modelling: Building shared understanding across stakeholders; Examples from natural resource management in Thailand
- **Learning Objectives:**
 - Diversity of approaches to knowledge co-production and collaborative learning; Facilitation skills for multi-stakeholder engagement and research ethics.
 - Role play games as learning tools to reveal system dynamics and navigate complexity

Session 9: Connecting Field to Theory (1 hour)

- **Content:**
 - Seminar-style lecture provided by JC Castella about "Theory U" applied to Nan Province context to facilitate a transformation in the student's perspective, helping them to integrate the complexity of the socio-ecological systems
 - "Theory U" serves as a conceptual bridge between theoretical reflection and field experience. It addresses political ecology dimensions observed in Nan; How do local decisions connect to broader patterns; Methodological reflections on experiential learning; Power dynamics and ethics in participatory research.
- **Learning Objectives:**
 - Synthesis of experiential learning: It enables a transition from mere description of facts toward a deeper understanding of the transformations at work.
 - Connection between the local and the global: Theory U provides a framework for analyzing how decisions made at the local level (by farmers or communities in Nan) are interconnected with broader structural patterns and global dynamics.
 - Reflection on the dynamics of change: It helps explore power relations, the ethics of participatory research, and the ways in which agroecological transitions actually unfold.

Week 4: Community Based Solutions and Agroecology Food Systems (4 hours)

- **Pre-reading:** Selected documents on learning centers and community businesses in Nan Province
- **Assignment:** Students organized in group around a limited number of community-based solutions (2 pages)

Session 10: Business Models in Sustainable Agriculture (1.5 hours)

- **Lead:** CU-SAR + NCC
- **Content:**
 - Supply chain thinking; Market-led production planning; Upstream-midstream-downstream integration

- Illustration of advanced applications and action planning in the context of Nan Province (e.g. Originan, PGS network, Nan Ken-Pa Laeo, others)
- **Learning Objectives:**
 - Understand agroecology economics; Develop viable business models; Connect sustainability and profitability; Implement community-based solutions
 - Analyze food systems holistically; Design integrated solutions; Ensure continuity across system components; Balance stakeholder needs

Session 11: Food Systems and Business Models (1.5 hours)

- **Lead:** CU-SAR + RMUTL-Nan + NCC
- **Content:**
 - Local case study introduction to students
 - Provide information on case studies and transformative action research
- **Learning Objectives:**
 - Understand business clusters, multi-actors' platform, forum challenges
 - Experience agroecological enterprises; Analyze business models in context; Network with practitioners

Session 12: Nan Field Trip Preparation (1 hour)

- **Lead:** CU-SAR + RMUTL-Nan + NCC
- **Content:**
 - Local case study introduction by group of students
 - Student-led presentations of their understanding of the local contexts and Q&A with local lecturers in Nan
- **Learning Objectives:**
 - Articulate integrated understanding; Demonstrate learning outcomes; Plan field activities
 - Group collaboration, team building

Phase 2: Intensive Field Block in Nan Province (29 hours over 7 days)

Location: Based in Nan Province with daily excursions to multiple learning sites

Coordination: NCC + RMUTL-Nan (logistics), CU (facilitation and participatory methods), KU (academic oversight)

Pre-field materials:

- Video tours of field sites and video introductions from farmer hosts
- Support documents on learning centers, statistics of Nan agriculture, role play scenarios we'll use in field, scientific literature, etc.
- Student journal template

Daily Rhythm:

- 07:00-08:00: Morning preparation
- 08:00-12:00: Field activities (farm visits, observations, hands-on work)
- 12:00-13:30: Lunch
- 13:30-17:00: Field activities (continued field work, meetings, processing)
- 17:00-18:30: Free time (rest, personal reflection, individual journaling)
- 18:30-20:00: Dinner
- 20:00-21:30: Evening seminars and group reflection

Note: On Days 5 and 6, the schedule is adjusted to accommodate extended role play game sessions (3 hours each)

Day 1 (Sunday): Arrival, Welcome and Team Building**Arrival and Orientation**

- Transportation to Nan, accommodation check-in
- Welcome from organizers
- Safety briefing and accessibility accommodations

Evening Welcome Gathering (17:00-18:30)

- Introduction circle: students, faculty, farmers, community partners
- Overview of week's activities and learning objectives
- Introduction to field journals and documentation methods

Shared Meal (18:30-20:00)

- Dinner with all participants
- Informal introductions and relationship building
- Small games to get to know each other

Learning Focus: Building trust, orienting to place, establishing learning community, introducing participatory tools

Day 2 (Monday): Rice-Based Agroecological Systems

Site: Rice farming communities (organic farmers in Nan Province) – Field and farm scales

Morning Field Activities (8:00-12:00)**8:00-10:00: Farmer-Led Farm Walk (Facilitated by: Local organic rice farmers + agronomist)**

- Walk through rice paddies with farmer guides
- Observe current system: varieties, spacing, water management, biodiversity
- Hands-on soil observation exercise: Soil texture, color, structure analysis; Soil organisms identification; Farmer knowledge of soil health indicators

10:00-12:00: Learning from Farmer Experience (Students work alongside farmers in ongoing tasks (weeding, water management, pest monitoring))

- Farmer shares history of this farm and transitions attempted
- Discussion of challenges faced: markets, labor, knowledge gaps, policy
- Observation exercise: Document what you see, hear, smell (all senses)

12:00-13:30: Lunch with Farmers

- Small groups (3-4 students) eat with different farming families
- Informal conversation about daily life, farming decisions, challenges, hopes

Afternoon Field Activities (13:30-17:00)

- Continuation of farm work alongside farmers
- Students rotate to different farming activities
- Individual observation and documentation time

Free Time (17:00-18:30) Rest, shower, prepare for evening; Personal reflection and journaling

Dinner (18:30-20:00)

Evening Seminar (20:00-21:30) - Facilitated by: CU + NCC + local facilitators

- Small group debriefs: What did you observe? What surprised you?
- **Introduction to companion modeling concepts applied to rice systems**
 - What are the key elements of the rice farming system we observed?
 - Who are the actors involved? (farmers, traders, consumers, government, etc.)
 - What are the relationships between elements?
- **Activity:** Begin mapping the rice system together (large paper, markers)
 - Draw the farm system as we understand it so far
 - Identify flows: water, nutrients, labor, money, information
 - Note tensions and trade-offs observed
- Preview tomorrow's companion modelling exercise with farmers; Mark what we don't yet understand with question marks

Learning Outcomes: Observation skills, ecological principles in practice, respect for farmer knowledge, introduction to participatory modeling

Day 3 (Tuesday): Rice Systems and Companion Modelling with Farmers

Site: Continuation at rice field and farm scales + market visit

Morning (8:00-11:00) - Companion Modelling Exercise with Farmers - Facilitated by: CU + local facilitators

- **Co-construction of shared model:**
 - Students present their understanding from yesterday: system mapping with farmers present

- Farmers add their perspectives: what's missing? What's important?
- Farmers and local experts explain their mental models of the system
- **Facilitated discussion:** What scenarios for improvement? How would an agroecological system look like?
 - Group works to integrate perspectives into shared diagram
 - What are the constraints to achieving it?
 - Where do different stakeholders have different goals or understanding?

Mid-morning to Lunch (11:00-13:30) - Market Visit: Tracing Rice from Farm to Consumer

- Visit local market where farmers sell agricultural products; Observe pricing, packaging, consumer interactions.
- Interview market vendors about food sourcing and quality
- Discussion: How do market demand shapes farming decisions? How to add market actors and relationships to our agroecological model?
- Lunch close to the market

Afternoon (13:30-17:00) - Case Study: Khem Pa Laeo Rice Field Ecology with Tourism (or other case study sites depending on the groups)

- **Site:** Ban Na Sam Kha / Khem Pa Laeo
- **Instructor:** Khem Pa Laeo and team (2 people)
- **Focus:** Integration of rice farming with tourism and ecosystem services
- **Activities:**
 - Farm tour highlighting biodiversity conservation
 - Discussion of diversified income strategies
 - Analysis of how tourism affects farming practices
- **Systems analysis:** How does tourism change the farming system model? Students conduct short interviews about economic outcomes

Free Time (17:00-18:30)

Dinner (18:30-20:00)

Evening Seminar (20:00-21:30) - Food Systems Analysis and Community-Based Solutions – CU-SAR

- **Content:**
 - Connect field observations to broader food system concepts (global and local)
 - Food security dimensions and consumer-producer relationships
 - Group activity: Create value chain diagram for rice (or other agricultural products) from today's case
- **Reflection:** How did the companion modelling and community-based solution processes change your understanding compared to just observing?

Learning Outcomes: Companion modelling skills, co-construction of knowledge, systems thinking, market linkages, economic-ecological trade-offs

Day 4 (Wednesday): Diversification, Agroforestry and Community Development

Site: Upstream Nan Community Natural Agriculture Center, or other learning centers or community enterprises

Full Day Case Study: Sustainability in Soil, Water, Forest, People and Environment

- **Lead Instructor:** Local practitioner and team (2 people)
- **Partner facilitation:** RMUTL-Nan

Morning Activities (8:00-12:00)

8:00-9:30: Farm Walk Through Diversified Production

- Fruit trees, vegetables, livestock integration
- Forest-agriculture-livestock integration
- Water management systems and soil conservation practices

9:30-12:00: Hands-On Participation about Agroecology Practice

- Survey farmers about value-added processing, e.g. fermented liquid fertilizer production, herbal products, etc. (RMUTL expertise)
- Learn crop rotation, intercropping, resource recycling techniques
- Facilitated discussion on risk, resilience, and innovation; What experiments worked/failed? Economic viability and market strategies

12:00-13:30: Lunch

Afternoon (13:30-17:00)

Group Work - Agroecosystem Analysis and Adaptive Management

- Discussion of observation-based decision making
- Small groups synthesize observations using agroecological framework
- Prepare presentations on: How ecological goals and economic goals interact in this system; What makes this system "agroecological"?; What are vulnerabilities and strengths?

Free Time (17:00-18:30)

Dinner (18:30-20:00)

Evening Seminar (20:00-21:30)

Group Presentations and Feedback

- Each small group presents 10-minute analysis
- Presentations shaped around roleplay games or community-based solutions
- Reflection on collective learning process: How did direct participation change understanding?

Learning Outcomes: Diversification strategies, farmer innovation, integrated system design

Day 5 (Thursday): Nan Forum - Multi-Stakeholder Perspectives and Role Play Game

Morning (8:00-12:00)

8:00-10:00: Local Government and Extension Perspectives

- **Guest speakers:** Local agricultural office, extension officers
- **Topics:**
 - Agricultural policy landscape in Nan
 - Government programs supporting sustainable agriculture and farmer innovation
 - Tensions between production goals and sustainability
- **Q&A focused on:** What would help bridge farmer needs and policy goals?

10:00-12:00: Student-Led Farmer Interviews

- **Activity:** Small groups conduct semi-structured interviews with 3-4 different "farmer types": Conventional farmer, Transitioning farmer, Innovative agroecological farmer, Part-time farmer with off-farm income
- **Guiding questions:** What are your main challenges? How do you make decisions about practices? What would help you farm more sustainably? What are your hopes for the future?
- **Additional focus:** Gather information for evening role play preparation

12:00-13:30: Lunch

Afternoon (13:30-17:00) - Transdisciplinary Processes (Group work)

- **Lead:** KU + CU + NCC (collaborative facilitation)
- **Content:**
 - Analysis of interview data (qualitative synthesis)
 - Sustainability assessment
 - Identifying environmental, economic, social trade-offs

Role Play Game Preparation (1 hour)

- Introduction to evening's role play scenario
- Review of roles and objectives
- Time for students to read and prepare their roles

Free Time (17:00-18:30) Rest and final preparation for role play; Review role cards and strategy

Dinner (18:30-19:30) *Note: Earlier dinner to accommodate extended role play session*

Evening Role Play Game Session (19:30-22:30)

Role Play Game: Farmer Decision-Making - Based on real situations students encountered during interviews

- Students receive final briefing on their role card with specific goals, constraints, and information
- Roles include: Different farmer types (conventional, transitioning, innovative, part-time); Local agricultural officer; Rice trader/middleman; Extension worker; Consumer representative; Bank/credit provider; Community leader; NGO representative
- **Scenario:** The community must decide on a proposal for transitioning to more sustainable agricultural production system

- **Game Play (2 hours):**
 - Round 1 (20 min): Individual planning - What are your goals and concerns? What information do you have/lack? What's your initial strategy?
 - Round 2 (60 min): Negotiation phase - Stakeholders meet in various configurations; Bilateral negotiations; Small group discussions; Coalition building; Try to find solutions or compromises
 - Round 3 (30 min): Community decision-making - Community meeting format; Presentations of proposals; Debate and discussion; Voting or consensus-building on decision
 - Observer notes (10 min): Observers (faculty, farmers) share what they noticed
- **Key pressures built into game:** Yield vs. sustainability tensions; Market access vs. autonomy; Time investment and opportunity costs; Risk and uncertainty; Information asymmetries between actors; Credit and capital constraints; Policy incentives and disincentives; Power imbalances between stakeholders
- **Debrief Session (45 min):**
 - **Breaking character:** Students step out of roles, share how they felt
 - **What emerged?** What tensions surfaced? What surprised you? What coalitions formed? Why?
 - **Systemic analysis:** What systemic barriers did you encounter? How do individual incentives conflict with collective goals? What would need to change to enable better outcomes?
 - **Connecting to reality:** What does this reveal about real farmer decision-making? How does this connect to the systems we've been studying? What did we learn that interviews alone wouldn't reveal?
 - **Methodology reflection:** What did you learn from being in role? How did embodying a perspective change your understanding?
 - **Implications:** What would help bridge these divides in real life? What role could participatory processes play?

Learning Outcomes: Multiple perspectives, policy-practice gaps, systems dynamics, empathy for different stakeholder positions, understanding structural constraints, experiential understanding of decision-making pressures

Day 6 (Friday): Business Models and Actor Networks

Morning Field Visits (8:00-12:00)

8:00-9:30: Originan - Agroecological Enterprise

- **Site:** Successful agroecological business with market outlets
- **Focus:** Business model analysis - Product development and branding; Market access strategies (cooperatives, direct sales, online); Quality assurance and certification; Balancing sustainability and profitability

9:30-11:00: PGS (Participatory Guarantee System) Network

- **Site:** Local PGS group meeting or community enterprise visit
- **Topics:** Alternative certification for organic/agroecological products; Farmer-to-farmer learning and quality control; Building trust with consumers; Network governance and decision-making

11:00-12:00: Local Farmer Network Presentation

- **Presenter:** Local farmer network representative
- **Content:** Collective strategies for marketing and knowledge sharing; Successes and challenges in building farmer cooperation-networks; Role of universities and external partners in supporting farmer initiatives

12:00-13:30: Lunch

Afternoon: Supply Chain Analysis and Game (13:30-17:00)

Introduction to Supply Chain Thinking (30 min) - CU-SAR

- **Content:** Supply chain thinking: downstream (market) to upstream (production); Ensuring all parts of system work continuously and consistently; Value chain development for agroecological products; Brief case study analysis from morning visits

Supply Chain Game for Agroecological Products (3 hours) – CU-SAR + CU FS

Setup and Introduction (20 min) - Classic Game adapted to agroecological product supply chain; Explanation of rules and roles; Distribution of materials

- **Participants play roles:** Farmer/Producer; Cooperative/Aggregator; Processor; Retailer; Consumer (simulated demand)

Game Play (100 min)

- **Rounds:** Each round represents one production/sale cycle
- **Mechanics:** Players make decisions about ordering and production quantities; Orders take time to be filled; Each actor only sees their own inventory and orders from their immediate customer; Demand varies

Comprehensive Debrief (40 min)

- **What happened?** Graph the inventory patterns - what do you see? When did things start to go wrong? Why? How did it feel at different stages of the game?
- **Understanding system behavior:** How did delays contribute to problems? What role did information gaps play? How did local decisions create system-wide problems?
- **Real-world connections:** How does this relate to farmer challenges we've heard about? What examples did we see in Originan or PGS networks? How do perishability and seasonality amplify these dynamics?
- **Solutions and strategies:** What would have helped (information sharing, communication, coordination)? How might cooperative structures or different governance address these dynamics? What are the implications for agroecological market development? Role of transparency and trust in supply chains

- **Systems thinking lessons:** What did this teach us about feedback loops? How does the game illustrate emergence and unintended consequences? Connection to rice systems and diversified farms we've studied

Free Time (17:00-18:30)

Dinner (18:30-20:00)

Evening Seminar (20:00-21:30)

Group Work: Developing Agroecological Transition Scenarios

- Small groups (4-5 students) choose one community studied
- Develop realistic transition scenario using insights from role play games: Current situation analysis (informed by farmer decision-making game from Day 5); Desired future state; Steps needed to get there (considering supply chain dynamics); Resources, knowledge, policies required; Potential obstacles (systemic barriers revealed in role plays); How to address information, coordination, and incentive challenges; Governance and institutional arrangements needed
- Prepare 15-minute presentation for stakeholder panel tomorrow including visual models/diagrams

Learning Outcomes: Economic viability of agroecology, network strategies, business model design, supply chain dynamics, systems behavior, importance of information flow and coordination, understanding of feedback loops and delays

Day 7 (Saturday): Synthesis, Evaluation and Celebration

Morning (8:00-12:00) Student Presentation to Stakeholder Panel and Synthesis

- **Audience:** 15-20 local farmers, stakeholders, facilitators, faculty from all institutions
- **Format:** Each group presents 15-minute scenario including: Visual model/diagram of proposed transition; Key actors and their roles; Systemic challenges identified through role play games; Proposed solutions considering supply chain dynamics; How to address coordination, information, and incentive problems
- **Stakeholders provide feedback:** What's realistic? What's missing? What would help? What have you tried that students didn't mention? Do the role play insights match your reality? Open discussion on each scenario
- **Collective synthesis:** Building a comprehensive model of agricultural transition in Nan - Integrate insights from all week's activities: Rice system models from Day 2-3; Diversified systems from Day 4; Stakeholder perspectives from Day 5 role play; Supply chain dynamics from Supply Chain Game (Day 6); Business models from field visits
- **Critical thinking:** Key leverage points for change; Systemic barriers that keep recurring (revealed through games); Information gaps that create problems; Knowledge gaps that remain; Questions for future research or action

12:00-13:30: Lunch

Afternoon (13:30-17:00) Student Exchange in Final Reflection Circle

- **Reflection questions:** How has your understanding evolved through participatory processes? What did companion modeling and role plays reveal that interviews alone couldn't? How did "playing" these systems change your understanding? How might these tools be useful in your future work?
- **Students share experiences** from their home contexts; How do challenges differ across regions? What participatory methods might work in different contexts? What would you adapt from the games for your own context?
- **Faculty and facilitators'** reflections on the week: What worked well this year? What surprised facilitators? Evolution of student understanding through games; Reflections on participatory methods: what worked, what was challenging? Discussion of "aha moments" from role play games and other moments
- **Evaluation questionnaire**

Free Time (17:00-18:30)

- Rest and prepare for celebration dinner

Celebration Dinner (18:30-20:00)

- Certificate distribution, special meal, and celebration

Learning Outcomes: Synthesis of learning, receiving feedback, building lasting relationships, practical skills in facilitation

PART C: ASSESSMENT & GRADING

Total Assessment: Three components weighted as follows

Component 1: Field Journal & Documentation (40%)

During Field Block:

- **Daily journal entries** with: Detailed observations; Sketches, diagrams, maps; Documentation of companion modeling processes (photos, sketches of models created)
- **Personal reflections;** Quotes from farmers and other participants; Questions and uncertainties; Reflections on role play games: Day 5: Farmer decision-making role play - your role, key moments, tensions, insights; Day 6: Game - what happened, emotional responses, systems

Submission (Due 1 week after field block): 3 to 4 pages of journal entries

- Documentation of a companion modeling session; reflection on both role play game experiences (Day 5 and Day 6); Systems thinking development
- How did participatory methods change your understanding? Which method was most powerful for you and why?

Grading Criteria:

- Depth and detail of observations (not just descriptions); Evidence of perspective shifts and learning; Engagement with farmer perspectives and knowledge

- Documentation of participatory processes and insights gained; Evidence of systems thinking development; Ability to connect game insights to real-world complexity

Component 2: Reflective Analysis Essay (40%)

Five to ten pages essay on ONE of the following topics:

- A. System Analysis:** Analyze one farming or entrepreneurial system you studied using an agroecological lens. What makes it agroecological (or not)? What ecological, economic, and social factors interact? What would successful transitions require? Include insights from companion modeling or role play experiences.
- B. Participatory Methods:** Reflect on your experience with farmers and communities. What did participatory methods reveal about agricultural systems that other approaches might miss? When and how might you use these tools in future work? What are the challenges and limitations of participatory approaches?
- C. Role Play Games as Learning Tools:** Compare and contrast the role play game experiences during Field Block. What did they reveal about agricultural systems? How did embodying different perspectives change your understanding? What are the strengths and limitations of gaming as a methodology for understanding complex systems?

Due: 2 weeks after field block

Grading Criteria:

- Clear argument and analytical depth; Integration of field observations with agroecological concepts
- Critical thinking (not just description); Evidence of personal learning
- Thoughtful reflection on methodological approaches showing understanding of different methods purposes and designs

Component 3: Collaborative Learning Contribution (20%)

During Field Block:

- Quality of participation and intellectual openness; Engagement with farmers as knowledge holders (respectful, curious, reciprocal); Contribution to group problem-solving and learning
- Participation in companion modeling and role play activities; Contribution to debriefs and analysis; Support of peers and collaborative spirit

Assessment Methods:

- Facilitator observations note on role play participation
- Self-assessment reflection (submitted with field journal)

Grading Criteria:

- Genuine, respectful engagement vs. performative participation; Meaningful questions and active listening
- Authentic engagement in participatory methods; Quality of contribution to debriefing sessions